Energy efficiency gain
Inference time change
Throughput decrease
ShareGPT
Llama3-8B
MATH
Llama3-8B
Natural Questions
Llama3-8B
ShareGPT
Qwen3-8B
MATH
Qwen3-8B
Natural Questions
Qwen3-8B
ShareGPT
Llama3-70B
MATH
Llama3-70B
Natural Questions
Llama3-70B
Change Rate
Batch size
64
128
256
512
40
30
20
10
0
−10
27.5
-1.6
-1.6
26.9
-1.4
-1.4
26.5
-0.5
-0.5
29.7
-4.9
-5.2
24.2
4.7
4.4
26.7
-0.2
-0.2
20.8
3.0
2.9
27.8
-1.4
-1.4
26.4
-2.4
-2.5
14.1
1.7
1.7
31.1
4.2
4.1
36.3
-1.3
-1.3
34.7
-1.4
-1.4
40.6
-7.8
-8.5
32.9
0.4
0.4
29.6
2.6
2.6
32.9
1.2
1.2
11.8
3.4
3.3
29.2
2.0
2.0
29.5
1.9
1.9
31.3
3.0
2.9
28.9
2.7
2.6
28.6
1.7
1.7
30.7
-0.6
-0.6
30.0
-0.1
-0.1
29.4
0.6
0.6
31.8
0.3
0.3